GEM-MM: Entropy-Guided Preference Alignment for Vision-Language Models

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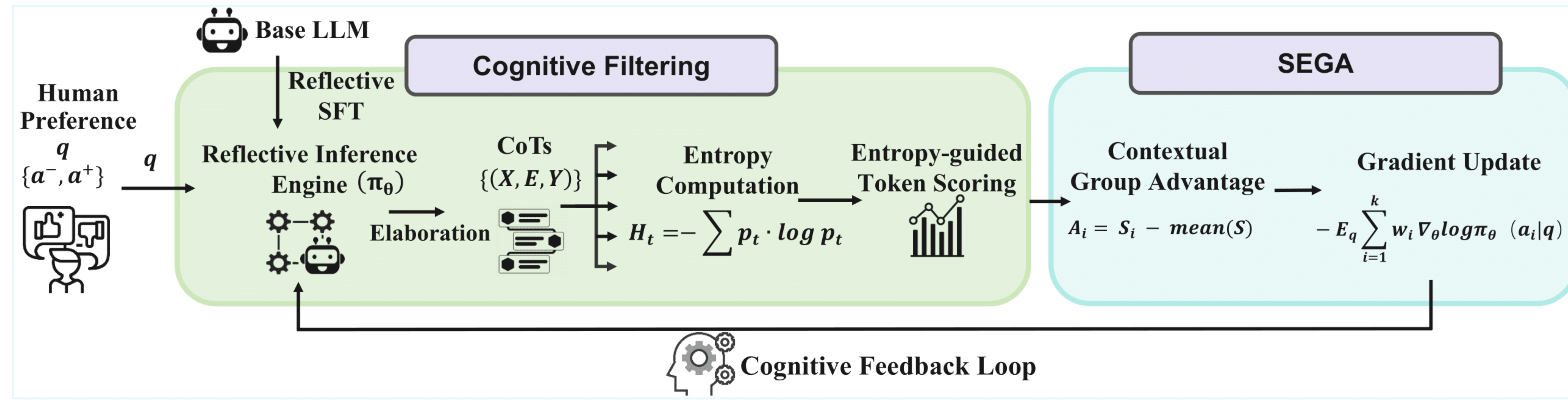
AAAI Draft

Full fine-tune ů Qwen3-VL-4B

github.com/SNOWTEAM2023/GEM-MM

Motivation & Approach



GEM closed-loop pipeline for VLMs (sample → entropy filter → SEGA update).

- Multimodal preference data is scarce, so alignment often uses only a few thousand pairs.
- Under a **locked 3000-prompt** MM-RLHF budget, MM-DPO, mDPO, and SIMA stay within **1.1** preference points of each other.
- Pairwise losses treat every token equally. We call this **entropy-blind preference optimization**.
- **GEM-MM** scores on-policy CoT candidates with a fork/final entropy reward and updates the full VLM with group-normalized SEGA.

Key Formulas

1 Token entropy

$$H_t = -\sum_v p_t(v) \log p_t(v)$$

2 Fork / final reward

$$r(y) = -\bar{H}_{\text{final}} + \lambda \bar{H}_{\text{fork}}$$

λ=2.0, ρ=0.05 (primary)

3 SEGA update

$$A_i = \frac{r_i - \text{mean}(r)}{\text{std}(r) + \varepsilon}$$

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_i (-A_i \log \pi_\theta(y_i \mid x))$$

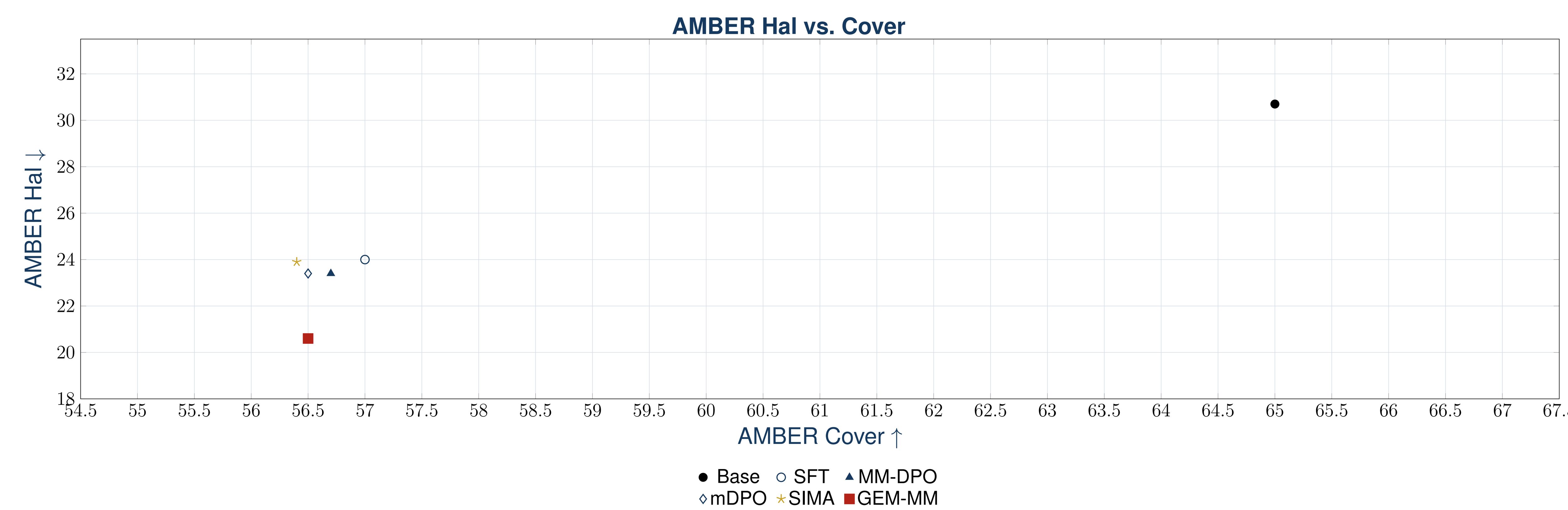
Main Results (MM-RLHF, fair 3000-prompt budget)

Preference & behaviour

Method	Depth Pref	Implicit	A/B	AMBER Hal↓
Base	51.2	—	**68.9**	30.7
SFT	57.9	40.9	66.4	24.0
MM-DPO	58.1	41.0	66.4	23.4
mDPO	58.4	40.2	66.1	23.4
SIMA	57.3	40.8	66.0	23.9
GEM-MM	**61.7**	**41.4**	**68.3**	**20.6**

CHAIR ↓: **4.6** (vs. 5.0–5.1). Cover matched at 56.4–57.0. Hallusion fAcc **45.95** (best). Conservative setting near-chosen **62.4%**.

Depth Pref shares form with the training reward (diagnostic). Implicit reward is objective-agnostic.



Takeaway. Pair construction saturates at small multimodal budgets. Reading fork vs. final entropy inside on-policy responses reduces generative hallucination at matched coverage while leading on preference depth.

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